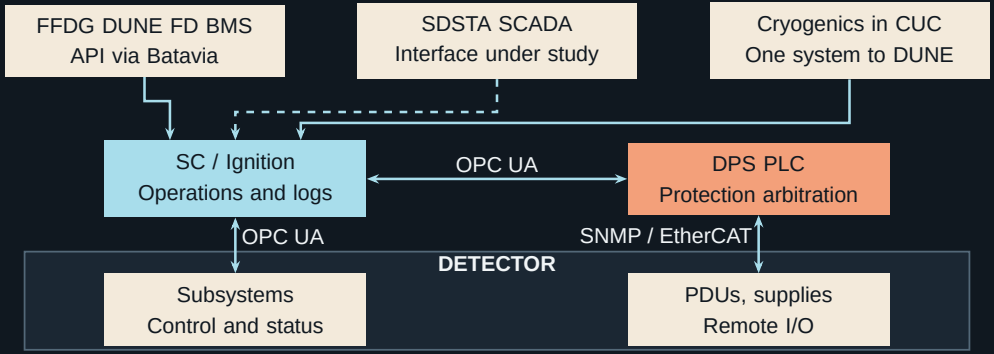
Definitions and responsibilities

DPS	Detector Protection System: arbitrates protection, applies permits and inhibits, and checks the response.
SC	Slow controls: operates and monitors equipment, coordinates requests and keeps the logs.
DAQ	Data acquisition: configures readout and records detector data.
BMS / cryo	Building management and cryogenics: operate their systems and report their state to SC.
Subsystems	Define operating limits, protection actions and allowed response times.
Local interlocks	Can trip equipment without SC or the central DPS.

Protection trips take priority over operating requests.

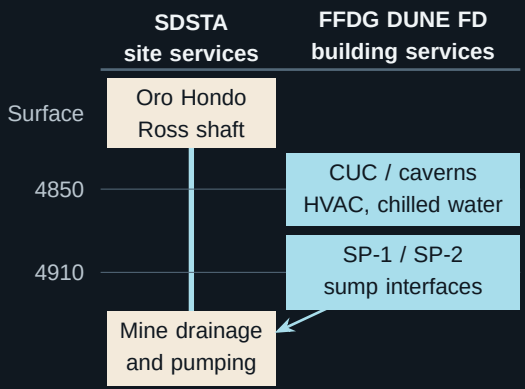
SDSTA issues personnel-safety alarms.

How the systems connect



OPC UA carries status and requests. DPS uses SNMP for power distribution units (PDUs) and supplies, and EtherCAT for remote I/O.

FFDG BMS: the building-services interface



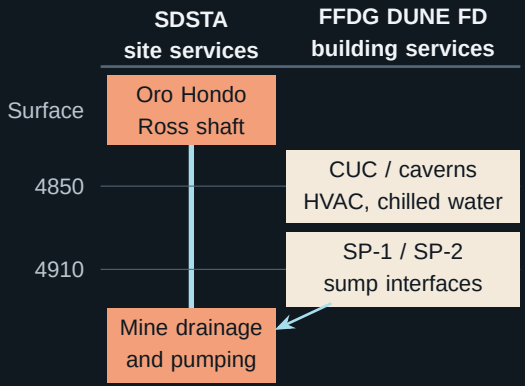
Proposed connection
ISD proposes a Metasys API in Batavia to connect the FFDG DUNE FD BMS to SC.

Tested with SBN building data
The API and OPC UA publication have been exercised.

Needed for live data
The Metasys API licence.

Service boundaries at SURF. The map shows the scope of the systems, not individual controllers.

SDSTA BMS: survey and interface status



Hardware survey underway
Locate controllers, input/output connections and network ports.

Protocols under investigation
Find which signals can be sent to SC.

Agreement under discussion
Set responsibilities for signal use, faults and maintenance.

SDSTA services extend beyond the detector caverns. The survey will establish the BMS connection points.

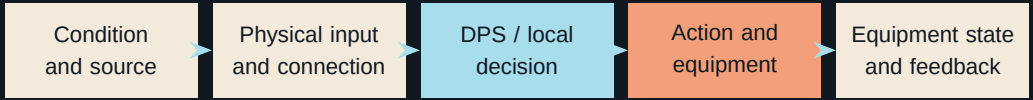
Cryogenics: one system to DUNE

The cryogenic controls are in the Central Utility Cavern (CUC). Their internal divisions appear as one system to DUNE.

Function	Detector interface
Process control	Cryogenics operates and protects its own process.
Protection inputs	Liquid-argon level, relief-valve state and boiling-risk conditions can affect HV and bias permits.
Information path	Cryogenics $\rightarrow$ SC $\leftrightarrow$ DPS. Trace each measurement to its actual sensor.
Before operation	Agree readiness states, limits and the response to missing or invalid data.

The detector running conditions are still being agreed.

Use the interlock matrix to review each system



Start with the interlocks proposed by the consortia in their interface documents (ICDs).

For each one, identify the existing hardware, missing connections and required additions. Check the time from detection to completed shutdown.

Use the rack lists, drawings, parts lists and power calculations.

DPS logic: allow, trip and recover



Allow: the request is valid, required conditions are met and no trip is active.

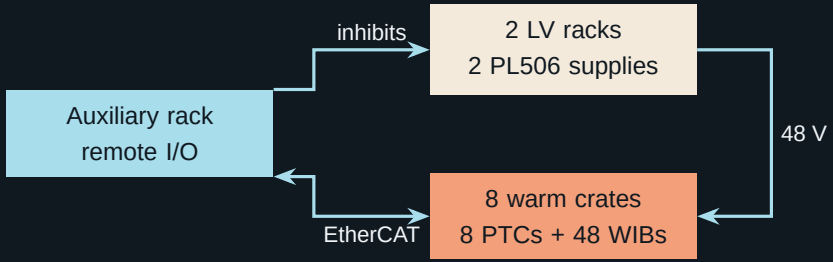
Trip: remove the permit, retain the cause and check the equipment response. SC logs the event.

Recover: reset the trip, then request a separate re-enable.

Proposed controller: a commercial programmable logic controller (PLC) based on BDE, located in the CUC. Loss-of-signal behaviour is set for each protected load.

Detector

BDE: the existing bottom-drift design

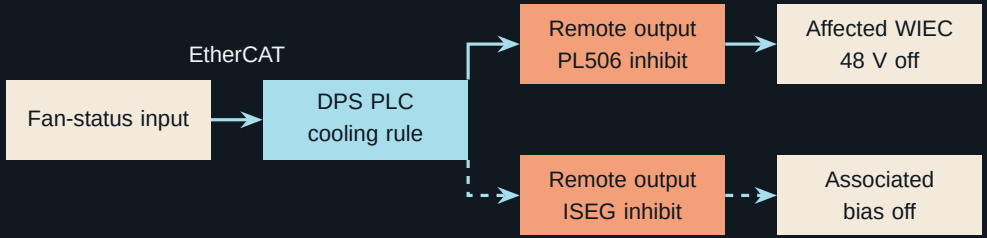


Ten groups: 10 auxiliary racks, 20 low-voltage racks and 20 PL506 supplies; 80 warm interface electronics crates (WIECs), containing 480 interface boards (WIBs).

Each crate has a power and timing card (PTC). Ten bias racks provide the separate charge-readout-plane and field-cage interfaces.

Available: panel drawings, parts lists, power calculations and Ignition screens.

BDE trip example: loss of fan cooling

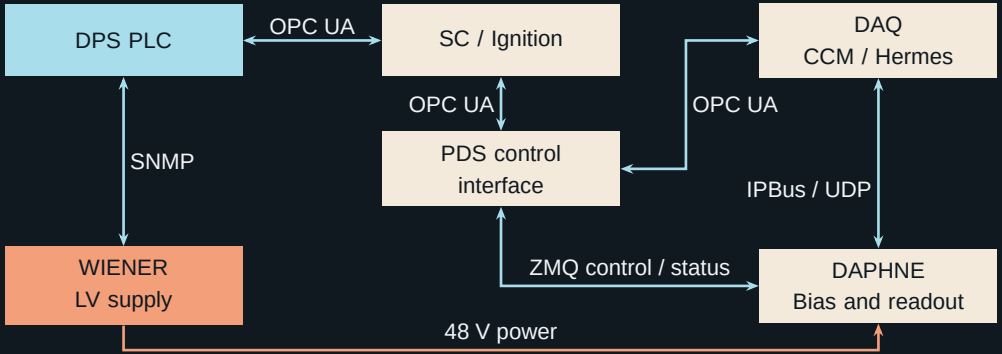


A cooling fault removes power from the affected crate. DPS checks the supply state and voltage; SC records the trip.

Solid path: documented fan-to-power-supply inhibit.

Dashed path: associated bias shutdown, if required by the VD interlock rule. The dependency and shutdown order are still being agreed.

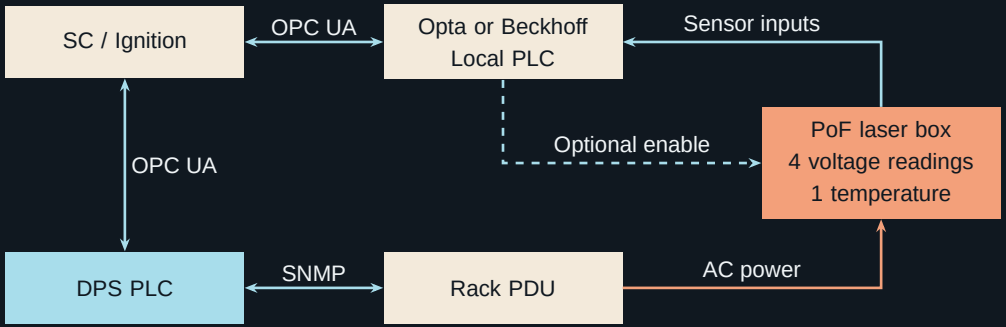
PDS DAPHNE: bias control, power and DAQ



Photon Detection System (PDS): 34 low-voltage feeds.

SiPM light-sensor bias inhibit: DPS → SC → DAPHNE. SC returns bias status and informs DAQ. Cathode-mounted devices use the PoF path on the next slide.

Power over fibre: monitor and switch off locally



80 boxes: 320 voltage + 80 temperature readings = 400 readings.

DPS inhibits power over fibre (PoF) through the PDU. A separate box-enable signal would provide another control option, if available.

High voltage: drift and detector bias

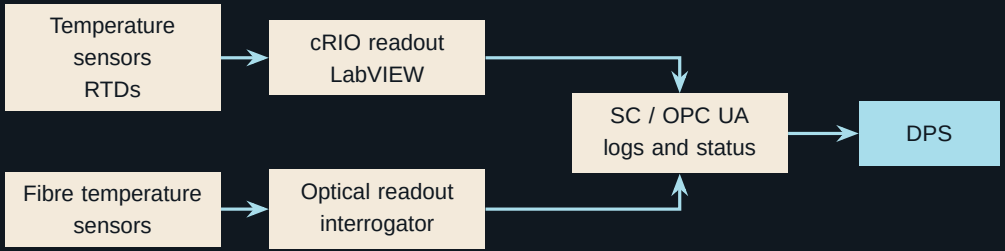
One drift-HV supply; ten bias crates with 576 channels.

480 channels serve the charge readout planes (CRPs); 96 serve the field cage.

Equipment	Control and monitoring	Protection
Heinzinger drift-HV supply	SC / OPC UA through LabVIEW and cRIO; analog control and readback	DPS remote I/O operates the wired interlock
ISEG bias supplies	DPS uses SNMP; SC receives status and logs	DPS remote I/O operates wired channel inhibits

Example: low liquid-argon level removes the drift-HV permit. A drift-HV trip also shuts down the associated field-cage termination supplies.

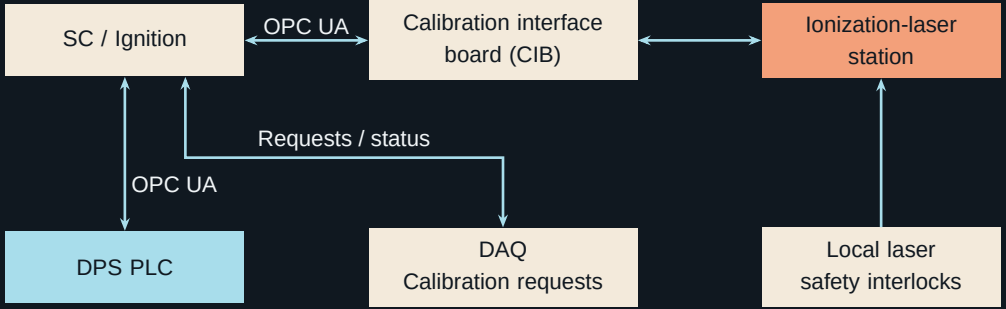
CalCI: temperature measurements



Calibration and Cryogenic Instrumentation (CalCI) measures conditions inside the detector. Temperature profiles guide cooldown and operation.

For protection, each measurement needs a location, a limit, an allowed response time and a named piece of equipment to shut down.

CalCI: ionization lasers



Four laser stations. Proposed rule: block conflicting laser and PDS requests. Restore PDS after confirmed emission-off or shutter closure.

Local laser-safety interlocks remain independent.

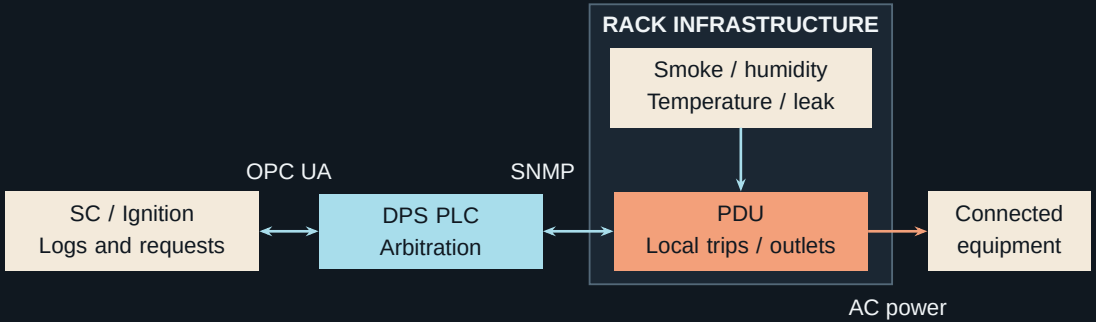
CalCI: purity monitors



Proposed approach: keep taking data and mark the intervals when the flash lamp is active.

Decision still needed: the older PDS interface document requires SiPM bias to be off during purity-monitor activity. Agree the rule before implementing the inhibit.

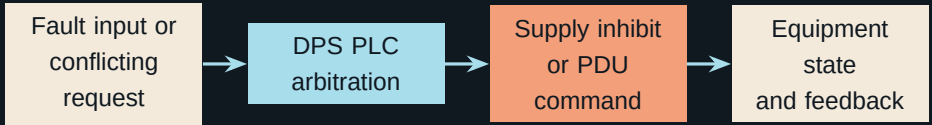
Rack infrastructure: shared protection



Rack protection groups the PDUs, sensors and connections. Local rules switch off affected outlets. Humidity is monitored; a humidity trip needs a separate requirement.

TDE + DAQ/timing: 336 microTCA crates. TDE uses 20 mezzanine racks. DPS controls their PDUs; crate-status and external-inhibit interfaces remain possible additions.

Test the complete protection path



Use the existing BDE test setup to measure detection through completed shutdown.

Test local trips, lost communications, reset and re-enable. Check that SC logs the cause, decision and final equipment state.

No critical function requiring less than 1 ms has been identified so far. The consortium review and hardware tests will check this assumption.

Hardware review and procurement: June 2027

Work before review	Result
Trace each inhibit	Existing hardware, additions, cable routes and quantities
Test representative paths	Measured trip times, recovery and SC logging
Prepare purchases	Panel space, power, drawings, parts, licences, costs and spares

Order after the readiness review; install to match the site schedule.

Facility interfaces may be needed in summer 2027. The full VD response matrix is due for review in early 2028.

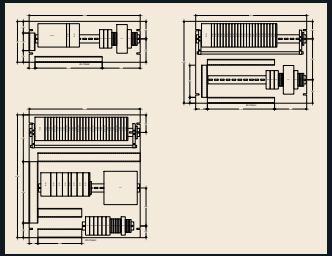
Who does the work

Proposed responsibility	Work
Manuel + control engineer	VD hardware design, SURF interfaces and procurement
Terri and Trevor	BDE panels, PLC and I/O implementation, bench tests
Geoff + Manuel	SC commands, DPS arbitration and recovery
Vivian + consortia / DPS	Equipment requirements and interface-document updates

Support needed: control-engineering time and access to the existing test stands.

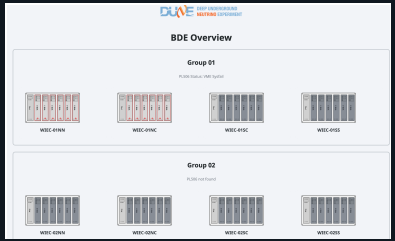
BDE reference: panels and operator screens

TPC protection panel designs



Central PLC, bias I/O and LV/fan/heater reference panels.

BDE Ignition screens

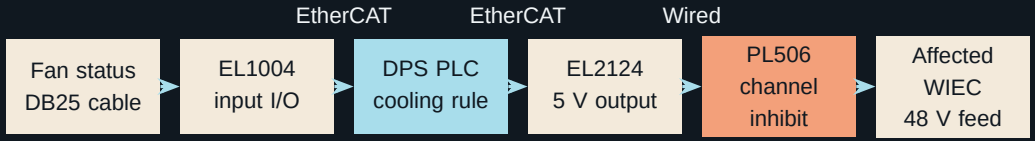


Existing Ignition display of the BDE crate groups.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

Reference panel design. The VD group contains eight warm crates.

BDE reference: the cooling-trip connections



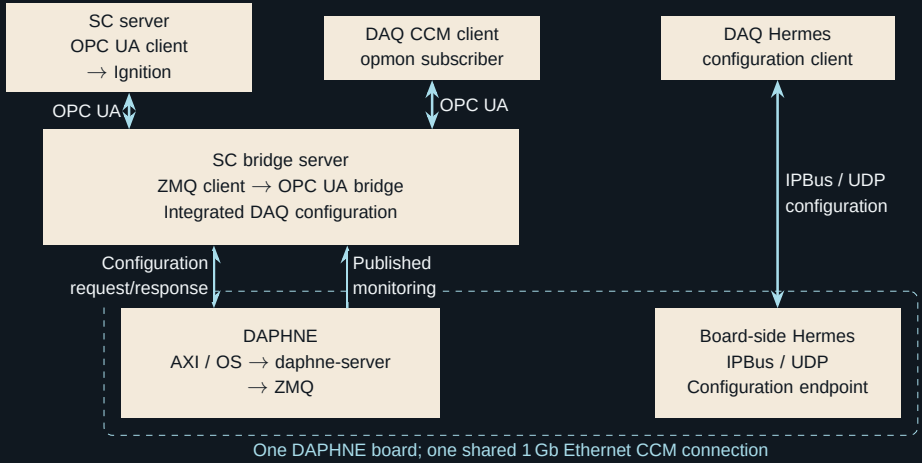
Cooling fault → withdraw the permit for the affected PL506 channel.

Feedback: PL506 state and voltage/current → SNMP → DPS; OPC UA carries the result to SC for logging.

VD allocation check: six fan-status signals per WIEC means 48 per eight-WIEC group. Match these to the populated VD inputs and cables.

Six fans per crate is the design case used here. These are existing input requirements, not additional I/O.

DAPHNE reference: control and monitoring services



Working SC–PDS–DPS ICD, Figure 1. DAQ CCM/opmon and Hermes remain distinct clients.

VD rack space and DPS allocation

Rack type	Count	DPS allocation and workbook space
BDE auxiliary	10	12U DPS; U3–U21 marked Space (19U); eight WIECs per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U DPS; U3–U12 marked Space (10U)

Space marked in the rack workbook is not a new DPS reservation. Free DIN-rail length and I/O capacity depend on the populated VD panel.

U denotes rack height. The marked space is not an allocation of free PLC terminals.

Bias-panel draft: two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

Draft layout: the difference between 1,440 outputs and 576 bias channels is not verified spare capacity.

VD power budget: component loads

Equipment	Power / capacity	Basis and limit
PL506 input	2.25 kW each	20 supplies: about 45.1 kW expected AC input
WIEC fans only	380 W per group	$8 \times 6 \times 0.33\,\text{A} \times 24\,\text{V}$; excludes heaters and I/O
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The inherited 831 W auxiliary figure needs a VD component budget covering fans, heaters and populated I/O.

Fan power assumes six fans per crate. Supply capacity is distinct from expected consumption.

Rack protection: reference hardware

Rack component	Reference hardware / allocation
PDU	Raritan PX3; inlet and outlet allocation for the connected loads
Smoke detector	4WTAR-B and EOLR-1 end-of-line relay
Contact interface	DX-PD2C5; 12 V supply to the smoke detector
Humidity	Record the probe and peripheral interface for each rack
Temperature / water leak	PDU peripheral sensors where fitted

The May workbook includes a 76-entry rack-protection/PDU allowance. Map the actual rack packages against this existing allowance.

Humidity probe models and any humidity-trip requirement remain to be specified.

Local PDU trips and recovery

PX3 setup guide	Local behaviour documented
Smoke, water, temperature	Local event rules and scripts can switch off the protected outlets
Network-switch outlet	Kept powered during the guide’s fallback shutdown
After a local trip	Power is restored deliberately; no automatic restoration

The VD control chain is Ignition → DPS PLC → PDU. Local trip status returns to the PLC and to SC for logging.

The reference prototype used direct Ignition control. The VD PLC path still needs its own communication-loss and recovery tests.

Other VD subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	34 LV feeds in the current presentation scope; supply status and control interfaces
Power over fibre (PoF)	80 boxes; 320 voltage and 80 temperature readings; Opta PLC or Beckhoff PLC alternative
TDE	20 mezzanine racks; DPS controls PDUs; crate-status and external-inhibit interfaces remain options
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

TDE and DAQ/timing contain 336 microTCA crates in total. Crate totals do not establish the number of PDUs or network switches.

BMS survey: 20 panel symbols in the drawing

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Marked inspection maps cover the 4850 and 4910 levels.

These are drawing references. The underground survey will establish what is installed.

VD rack inventory: May 2025 allocation

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

Rack positions include the reserved and spare allocations; they are not counts of PLCs or I/O stations.

Device hierarchy: controls, facilities and racks

SC and network
- Ignition and logging services
- Operator / engineering stations
- Switches, routers and firewalls

DPS
- Central PLC and 24 V supplies
- EtherCAT couplers / fibre links
- Digital / analog / temperature I/O
- Fan/heater power outputs
- Wired permit / inhibit outputs

Facility interfaces to SC

FFDG BMS
- Metasys / Batavia API
- Building controllers

SDSTA BMS
- Controllers and I/O: survey

Cryogenics
- Process and safety PLCs
- Operator stations / instruments

Rack infrastructure
- Raritan PX3 PDUs
- Smoke detectors and relays
- Contact interfaces / supplies
- Humidity / temperature probes
- Water-leak sensors

PDU-connected equipment
- Subsystem supplies and crates
- PoF boxes
- Computers and other rack loads

Facility systems keep their own controls. Rack protection and shared I/O are counted once.

Device hierarchy: readout electronics and HV

336 microTCA crates in total across TDE and DAQ/timing.

Bottom Drift Electronics
- 10 groups; 20 PL506 supplies
- 80 warm crates
  - 80 power/timing cards; 480 WIBs
  - Fans, heaters and temperature inputs

High voltage
- 1 Heinzinger supply; cRIO / LabVIEW
- 10 ISEG bias crates
  - 576 channels: 480 CRP + 96 field cage
- 10 bias racks with remote I/O

microTCA crate equipment
- Carrier hubs (MCHs)
- Readout / timing modules
- Power units and cooling fans

Top Drift Electronics
- 20 mezzanine racks
- 20 LV supplies and distribution units
- Front-end boards / calibration units
- White Rabbit timing switches

DPS control of TDE is through the PDUs.

Device hierarchy: PDS and DAQ / timing

Photon Detection System
- 42 DAPHNE boards in the working inventory
- WIENER supplies: 34 LV feeds

Power over fibre
- 80 laser boxes
  - 320 voltage + 80 temperature readings
- Opta or Beckhoff monitoring PLCs
- PDU power; optional box-enable input

Light calibration and timing
- 8 light-calibration boxes
- Local optical timing fanouts

DAQ and timing
- DAQ computers and readout switches
- Timing / readout microTCA crates
  - Included in the 336-crate total
- GPS interface boards (GIBs)
- GPS-disciplined oscillator
- Timing masters and fanouts

DAQ data and timing use separate links. SC receives the agreed status and activity information.

Device hierarchy: calibration and instrumentation

Temperature
- RTDs; readout crates / cards
- cRIO and LabVIEW stations
- Fibre sensors and interrogators

Liquid level
- CRP / field-cage sensors and readout boards
- Level transmitter and controller

Other instrument interfaces
- Cold cameras; GIZMO; impedance monitor
- Berghof / Siemens instrument controllers

Ionization lasers: 4 stations
- CIB and station controllers
- Power electronics and computers
- Shutters and local safety interlocks

Purity monitors
- HV/LV supplies and front-end electronics
- Flash lamps, lamp supplies and triggers
- Digitizers and LabVIEW computers

Pulsed neutron source
- Generator, controller and permit interface

The device list records the interfaces and quantities still awaiting allocation.